CFTC	Commodity Futures Trading Commission
CMO	collateralized mortgage obligation
CMS	Constant Maturity Swap
CMT	Constant Maturity Treasury
CSA	Credit Support Annex
CVA	Credit Valuation Adjustment
FAS	Financial Accounting Standards
FCM	futures commission merchant
FHA	Federal Housing Administration
FHLMC	Federal Home Loan Mortgage Corporation
FNMA	Federal National Mortgage Association
FV	future value
GC	general collateral
GSEs	government-sponsored enterprises
IANs	index amortization notes
IO	interest only
ISDA	International Swaps and Derivatives Association
JPY	Japanese yen
L	LIBOR
LBSF	Lehman Brothers Special Financing Inc.
LEANs	LIBOR Enhancement Accrual Notes
LIBOR	London Interbank Offered Rate
LTV	loan to value
MBS	mortgage-backed security
mm	million
NPV	net present value
OCC	Office of the Comptroller of the Currency
OIS	Overnight Index Swap
OTC	over-the-counter
P&L	profit and loss
PMT	annuity payment per period
PV	present value
PV01	present value of a basis point
qm	quarterly money; quarterly Act/360
SABR	Stochastic Alpha, Beta, Rho
sb	semi-bond; semi 30/360
SEC	Securities and Exchange Commission
SNs	structured notes
T	time to maturity or time to expiration
T_3	yield of the on-the-run 3-year Treasury
T_5	yield of the on-the-run 5-year Treasury

T_{30}	yield of the on-the-run 30-year Treasury
USD	U.S. dollars
vol	volatility
zc	zero coupon

Chapter 1

An Introduction to Swaps

The interest rate swaps market has experienced tremendous growth since what is commonly regarded as the first swap was executed in 1981. In that year Salomon Brothers intermediated a cross-currency swap between the World Bank and IBM in a transaction that at the time was unique and provided considerable advantage to both counterparties. The growth in the market since then manifests itself not only in the vast increase in the notional outstanding of interest rate swaps but also in the varied users and uses of swaps. The purpose of this chapter is to provide a broad overview of the swaps market. We will focus on products and conventions in the market.

1.1 Overview

Most discussions on swaps start with some analysis of the so-called comparative advantage argument: one counterparty has a relative advantage borrowing on, say, a fixed rate basis and another has a relative advantage borrowing on a floating rate basis. The argument then goes that it makes sense for these two guys to borrow funds in their relatively advantaged manner, then get together and swap their cash flows, exchanging fixed cash flows for floating ones, and vice versa, should they actually prefer to raise funds in their non-advantaged manner.

But let's leave it at that. This comparative advantage argument may very well have been the motivating factor behind the genesis of the swaps market over 30 years ago but doesn't play a huge role today.[1] Nowadays the swaps market is so liquid and vast that when an investor executes a swap, a dealer who takes the other side does not need to first line up yet another counterparty on the other side. These days if someone wants to do a swap, he approaches any one of a number of major swap dealers to do the trade.[2] And nearly as fast as one could execute a Treasury trade, one can now execute a swap.[3]

A good way to start looking at the swaps market is to see just how big it is and just how much it has grown over time. One measure of the size and growth of the swaps market is to look at how the outstanding *notional* (or fictitious principal) on all trades has changed from year to year. Figure 1.1 shows just how rapid this growth has been. The outstanding notional of interest rate swaps, interest rate options, and currency swaps combined first crossed the \$100 trillion threshold in the year 2003 and crossed \$400 trillion by 2008. By comparison, the outstanding amount of U.S. Treasury securities in 2008 was on the order of magnitude of \$5.7 trillion.[4] This is a bit of an apples and oranges comparison since the outstanding notional of derivatives relates more to trading volume (and does not account for the fact that some outstanding trades offset one another, meaning the net risk in the marketplace may be far less than suggested in Figure 1.1), and the outstanding

[1] Comparative advantage was indeed crucial to the swap arranged between the World Bank and IBM in 1981. At that time, IBM had outstanding debt denominated in Swiss francs and in Deutsche Marks. The World Bank had outstanding debt denominated in U.S. dollars through the issuance of a Eurodollar bond. The swap negotiated by Salomon had the World Bank assuming the debt payments of IBM in francs and Deutsche Marks and, simultaneously, had IBM assuming the debt payments of the World Bank in dollars.

[2] The swaps market is an *over-the-counter* (OTC) market in which trades are negotiated directly between counterparties and not through an exchange. Buy-side accounts will execute swaps directly with any of a number of dealers who are market makers in swaps. Brokers loosely connect dealers by helping to match offsetting dealer positions (on an anonymous basis), which promotes liquidity in the market. By the way, when we say that "someone" does a swap, we refer to entities such as corporations, banks, insurance companies, government-sponsored enterprises, asset managers, hedge funds, and so on. Very few individuals would be acceptable by dealers as a counterparty in a swap due to the counterparty credit risk inherent in the contracts, as we will discuss in Section 2.4.

[3] That assumes of course that one has a relationship with a swap dealer, which is typically a prerequisite for entering into a swap. Usually, this means that the two counterparties have an International Swaps and Derivatives Association (ISDA) Master Agreement (which is a legal document that governs terms of trade) in place prior to trading. Moreover, the two counterparties would have done their homework on one another and feel comfortable with the counterparty credit exposure that would result from entering into the swap. The financial crisis that began in 2007 had a severe impact on liquidity in the swaps market, as it did in most every market. At the height of the crisis, some counterparties refused to do trades with certain other counterparties out of concern over counterparty credit risk, and liquidity was poor or nonexistent. We will talk about these issues in Section 2.4.

[4] Source: http://www.sifma.org/research/statistics.aspx.

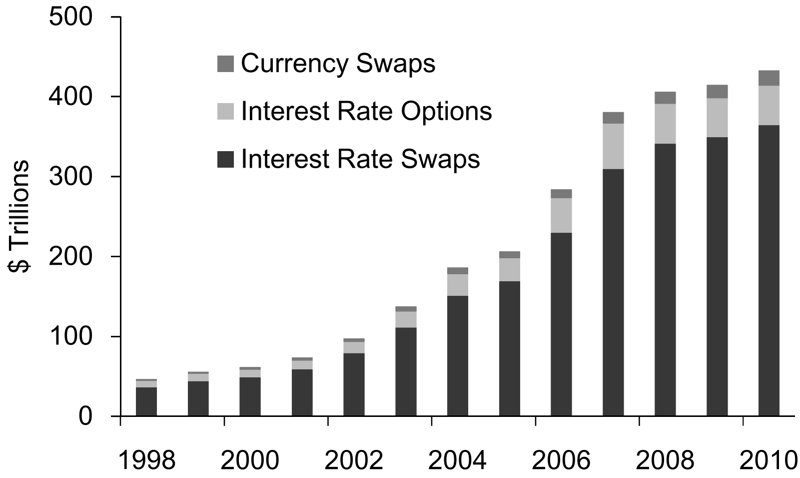


Figure 1.1: Growth of the Swaps Market. Amounts Outstanding of OTC Interest Rate Swaps, Interest Rate Options, and Currency Swaps
Source: Bank for International Settlements.

amount of Treasuries is not indicative of trading volume. But hopefully this leaves the impression that the swaps market is pretty big.

Although the market has standardized or "plain vanilla" trades, it has the advantage that its products can be completely tailor-made or customized to reflect almost any interest rate or currency outlook and/or to create any risk exposure profile. The traditional uses of swaps include liability hedging, balance sheet management, and asset hedging. The nontraditional uses include speculation, with both macro and relative value trading in the swaps market popular. That is, investors now use the swaps market itself to express views that they have in the market. In addition, the swaps market has become a benchmark by which the relative richness or cheapness of other asset classes (e.g., mortgages and agency debentures) can be ascertained and a vehicle through which views on these assets can be expressed.

1.2 Swaps

A *swap* is a contractual agreement between two counterparties that agree to exchange streams of payments over time. If both streams of payment are made in the same currency, then the trade is known as an *interest rate swap*; if the streams are made in two different currencies, then the swap is referred to as a *cross-currency swap*. We will spend virtually all of our time focusing on interest rate swaps, and we will further restrict our attention to swaps denominated in U.S. dollars (USD).

Interest rate swaps are subdivided into two subcategories: a *coupon swap* or *fixed-floating swap* or *plain vanilla swap* refers to a swap in which one stream is a fixed rate of interest, and the other is a floating rate of interest. In a *basis swap*, both streams are floating rates of interest.

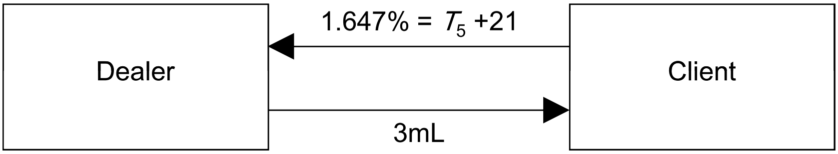
1.2.1 Fixed-Floating Swaps

The structure and terms of a plain vanilla swap with standard market conventions are shown in Figure 1.2. One counterparty, known as the client, pays a fixed rate of interest equal to 1.647% semiannually on a 30/360 (pronounced "thirty three-sixty") basis, and the other counterparty, known as the dealer, pays 3-month LIBOR, a floating rate of interest, quarterly on an Act/360 (pronounced "actual three-sixty") basis.[5] Instead of saying "30/360," people will sometimes say "bond," and instead of saying "Act/360,"

[5]LIBOR stands for London Interbank Offered Rate, a very common floating rate benchmark. In the USD swaps market, the standard is LIBOR as determined by the British Bankers' Association (BBA), sometimes abbreviated USD-LIBOR-BBA. Each day a number of contributor banks are individually asked by the BBA their perception as to where they could borrow funds for various maturities in the interbank market just prior to 11:00 A.M. in London. For each maturity, the top and bottom quartiles are tossed, and an average of the remaining levels is computed, representing the LIBOR quote for that day. Maturities vary, but are commonly 1, 3, 6, and 12 months.

LIBOR is a money market rate, and as such it is common that interest accrues on an Act/360 basis, meaning that interest is calculated based on the actual number of days in the accrual period. By comparison, if interest accrues on a 30/360 basis, then interest is calculated assuming that each month has exactly 30 days, regardless of the actual number of days in the month. For example, if interest is accrued between February 15 and March 15 of a given year, the number of days in the accrual period will be 30 if interest is calculated on a 30/360 basis. But if interest is accrued on an Act/360 basis, the number of days in the accrual period will be either 28 or 29, depending on whether or not it is a leap year.

LIBOR is the basic building block of interest rate swaps, a concept we shall discuss further in Chapter 3. The very integrity of LIBOR was called into question during the financial crisis. In an article appearing in the *Wall Street Journal* on May 29, 2008, titled "Study Casts Doubt on Key Rate: WSJ Analysis Suggests Banks May Have Reported Flawed Interest Data for Libor," it was suggested that a number of banks may have been significantly underestimating their costs of borrowing in the interbank market throughout the crisis. This would not only have served to present a healthier picture of the financial system than may have existed, but it also would have impacted the valuation of countless securities, contracts, mortgages, and so on, that depended on LIBOR. Subsequent studies discounted (no pun intended) the hypothesis that banks were underrepresenting their true borrowing costs (see, e.g., Gyntelberg and Wooldridge [2008] and González-Hermosillo, Mark Stone et al. [2008]), but some in the industry have remained skeptical.



Client pays:	1.647% (= T_5 + 21) semiannually on 30/360 basis
Dealer pays:	3-month LIBOR (“3mL”) quarterly on Act/360 basis
Notional:	\$100mm
Maturity:	5 years

Figure 1.2: A Plain Vanilla Swap

people will sometimes say “money.”[6] Oftentimes when people speak about swaps, they will use terms like this since they are quicker to say, and every second counts when the market is moving and it is desirable to execute a trade quickly. In fact, when referring to the payment frequency and day count convention in a standard swap, sometimes people will abbreviate this even further by saying something like “semi bond versus threes.” Saying “threes” surely takes less time than saying “3-month LIBOR paid quarterly on an Act/360 basis.” And for counterparties that deal with each other frequently and are comfortable that these standard conventions always apply when they transact swaps with one another, they may not even bother to discuss standard payment and day count conventions before doing a swap.[7]

The swap in Figure 1.2 has a maturity of five years and the *notional*, or fictitious principal, upon which payments are made is \$100mm.[8] Note that there will be a total of ten payments made by the client (the first of which is in six months) and 20 payments made by the dealer (the first of which is in three months) over the life of the swap. Final payments will occur five years after the *effective date*, or start, of the swap. The actual payments made on the fixed and floating side in the swap are

[6] The link between the word “bond” and “30/360” comes from the fact that many bonds pay coupons on a 30/360 basis. The link between the word “money” and “Act/360” comes from the fact that many money market instruments accrue interest on an Act/360 basis.

[7] It’s kind of like when you go into the same deli for lunch every day, and the guy behind the counter knows exactly what kind of sandwich you want without your having to say anything.

[8] The abbreviation “mm” stands for “million.” Thus, \$100mm stands for \$100,000,000. Similarly “bn” stands for “billion.” Thus, we will abbreviate \$1,000,000,000 by \$1bn.

$$\text{Fixed Payment} = \text{Fixed Rate} \times \text{Notional} \times (\text{Bond Days}/360)$$
$$\text{Floating Payment} = \text{3-month LIBOR} \times \text{Notional} \times (\text{Act}/360), \qquad (1.1)$$

where Bond Days is the number of bond days in the semiannual period (i.e., the number of days assuming interest is accrued on a 30/360 basis), and Act is the number of actual days in the quarterly period (see Appendix C for more details on day count conventions).

Example

Suppose the swap depicted in Figure 1.2, a \$100mm 5-year swap with a fixed coupon of 1.647% semi bond versus 3-month LIBOR quarterly Act/360, has an effective date of August 13, 2010, and a maturity of August 13, 2015. Of the ten semiannual periods on the fixed side, the last period is from February 13, 2015, to August 13, 2015. There are 180 bond days in this semiannual period, and thus the final fixed payment in the swap is

$$1.647\% \times \$100\text{mm} \times \frac{180}{360} = \$823,500.$$

Of the 20 quarterly periods on the floating side, the last period is from May 13, 2015, to August 13, 2015. There are a total of 92 actual days in this quarterly period. Denoting the final LIBOR setting in the swap by $x\%$, the final payment on the floating side of the swap is

$$x\% \times \$100\text{mm} \times \frac{92}{360},$$

a quantity that will not be known until the LIBOR setting date in May 2015. □

A standard swap has no upfront cash flows; that is, there is no "price" that is paid by either counterparty to get into the trade. The fixed rate is set when the trade is executed such that the swap has a net present value of zero (which we will denote NPV = 0).[9] Note that Figure 1.2 shows that the fixed rate in the swap, which is 1.647%, is also expressed as T_5+21, where T_5 represents the yield of the *on-the-run* 5-year Treasury at the time the swap is executed.[10] Thus the *swap rate* (also called the *par swap rate*), the fixed rate set

[9] Technically, it is incorrect to assert that a swap is an NPV = 0 transaction if there are bid/offer (i.e., transaction) costs in the swap. We will, however, ignore these costs for our purposes here and assume that unless otherwise stated, all swaps are NPV = 0 transactions when they are first executed.

[10] The *on-the-run* Treasury for a particular maturity is the most recently issued Treasury security of that maturity. Once a new issue has been issued, the issue that had previously been the on-the-run is referred to as the *old* or *single old*. The issue prior to that is referred to as the *double old*, and so on.

in the swap when it is first executed such that the trade has zero net present value to either side, is equal to the yield of the on-the-run 5-year Treasury plus some spread. In this case the spread is 21 basis points (bps).[11] The *swap spread* for a swap of a given maturity is defined as the spread in bps that is added to the yield of the on-the-run Treasury of comparable maturity to obtain the swap rate.[12] Thus in this example the 5-year swap spread is 21 bps.

Oftentimes when people talk about swaps and where swaps are trading, they just talk about swap spreads. It is implicitly understood that if you actually want to enter into a swap, the swap spread for a given maturity will be added to the yield of the on-the-run Treasury for that maturity to determine the swap rate. A swap rate has economic significance; it is the unique fixed rate in a swap that gives the swap zero value when the swap is first executed. A swap spread is just a benchmark that makes it easy to converse about swaps.[13] Since Treasury yields are easily observable by people in the market and also may be quite volatile from one moment to the next (certainly as compared to swap spreads), it is often easier to just talk about swap spreads until the trade is executed. For example, an investor who is interested in knowing where 5-year swap rates are offered (i.e., the rate at which a dealer is willing to receive fixed in a 5-year swap) might call up a dealer and ask for an *indication*, a non-executable market level that suggests where a transaction might be able to get done. The dealer might respond with either something like “1.647” or “21.” If the investor hears 21, then he knows that if he wants to pay fixed in the swap, he should expect to pay a rate that is around 21 bps higher than the yield of the on-the-run 5-year Treasury at the time the trade is executed (provided that the market for swap spreads doesn’t change, of course).

When we talk about a swap in which the floating side is 3-month LIBOR, we focus exclusively on the fixed rate in the swap. When an investor is receiving fixed in a swap, he wants to receive the highest fixed rate he can,

[11] A basis point (abbreviated bp and pronounced “bip” in the singular and abbreviated bps and pronounced “bips” in the plural) is one one-hundredth of one percent. Thus, a bp is 0.01%, or equivalently 0.0001. So if, for example, the rate 3.02% is increased by a bp, the new rate is 3.03%. If the rate of 3.03% is raised by one-quarter of a bp, the new rate is 3.0325%. In the swaps market, it is common to quote rates to a precision of a tenth of a bp, or alternatively, in quarter of a bp increments.

[12] Note, for example, that whereas a 5-year swap executed today has a maturity of five years from its effective date, the on-the-run 5-year Treasury will in general have a maturity of less than five years, since it was issued at some prior point in time. Nevertheless, it is customary in the market to quote a swap spread against the relevant on-the-run Treasury benchmark of comparable maturity.

[13] Actually, a swap spread is more than just a benchmark. Swap spreads are tradable themselves, with market participants taking views on the direction of swap spreads, up or down. We will discuss this concept further both in Chapter 2 and in Chapter 8.